2.3 Abstract Algebra II: Solutions

Warm-up

1.

Euclidean. N is multiplicative: $N(zw) = |zw|^2 = N(z)N(w)$. Let $a, b \in \mathbb{Z}[i]$, $b \neq 0$. In $\mathbb{C}$, $a/b = u + vi$ with $u, v \in \mathbb{Q}$. Choose integers m, n with $|u - m| \leq \frac{1}{2}$, $|v - n| \leq \frac{1}{2}$, and let $q = m + ni$, $r = a - qb \in \mathbb{Z}[i]$. Then

$$N(r) = |b|^2 \, |a/b - q|^2 \leq N(b) \left(\tfrac{1}{4} + \tfrac{1}{4}\right) < N(b).$$

Units. If $uv = 1$ then $N(u)N(v) = 1$, so $N(u) = 1$, i.e. $a^2 + b^2 = 1$: $u \in \{\pm 1, \pm i\}$. These are units.

Irreducibility via norms. If $N(z)$ is a prime number, z is irreducible: a factorisation $z = xy$ gives $N(x)N(y) = N(z)$, so one of $N(x), N(y)$ is 1.

- $2 = (1+i)(1-i) = -i(1+i)^2$. $N(1+i) = 2$ is prime, so $1+i$ is irreducible; $1-i = -i(1+i)$ is an associate.
- $5 = (2 + i)(2 - i)$, with $N(2 \pm i) = 5$ prime, so both factors are irreducible. They are not associates: the associates of $2 + i$ are $2 + i, -2 - i, -1 + 2i, 1 - 2i$.
- 3 is irreducible: if $3 = xy$ with non-units, $N(x)N(y) = 9$ forces $N(x) = 3$, but $a^2 + b^2 = 3$ has no integer solutions.

2.

Let $\varphi : \mathbb{Z}[x] \to \mathbb{F}_2$, $g \mapsto g(0) \bmod 2$. It is a surjective ring homomorphism. Its kernel contains $(2, x)$; conversely if $g(0)$ is even, $g = g(0) + xh(x) \in (2, x)$. So $\mathbb{Z}[x]/(2, x) \cong \mathbb{F}_2$, a field: $(2, x)$ is maximal (in particular proper).

Suppose $(2, x) = (f)$. Then $f \mid 2$, so f has degree 0, $f \in \{\pm 1, \pm 2\}$. Also $f \mid x$; but ± 2 does not divide x in $\mathbb{Z}[x]$ (the coefficient 1 is odd). So $f = \pm 1$ and $(2, x) = \mathbb{Z}[x]$, contradicting properness. So $\mathbb{Z}[x]$ is not a PID.

3.

(a) If $x^4 + 1 = g(x)h(x)$ non-trivially, then $(x + 1)^4 + 1 = g(x + 1)h(x + 1)$ non-trivially. But

$$(x + 1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$$

is Eisenstein at $p = 2$. So $x^4 + 1$ is irreducible.

(b) A cubic is reducible over $\mathbb{Q}$ iff it has a rational root. By the rational root test the only candidates are ± 1, and $f(1) = -3$, $f(-1) = 1$. So f is irreducible.

(c) Reduce mod 2: $\bar{f} = x^4 + x + 1 \in \mathbb{F}_2[x]$ has no roots ($\bar{f}(0) = \bar{f}(1) = 1$). The only irreducible quadratic over $\mathbb{F}_2$ is $x^2 + x + 1$ (the others, $x^2, x^2 + 1 = (x + 1)^2, x^2 + x$, have roots), and $(x^2 + x + 1)^2 = x^4 + x^2 + 1 \neq \bar{f}$. So $\bar{f}$ is irreducible. If $f = gh$ over $\mathbb{Q}$ with g, h of positive degree, by Gauss's lemma we can take $g, h \in \mathbb{Z}[x]$ monic, and then $\bar{f} = \bar{g}\bar{h}$ with the same degrees: contradiction.

4.

$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. If $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$, write $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$; squaring, $3 = a^2 + 2b^2 + 2ab\sqrt{2}$, so $ab = 0$. If $b = 0$, $a^2 = 3$; if $a = 0$, $2b^2 = 3$, i.e. $(2b)^2 = 6$; neither has a rational solution. So $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$ and by the tower law $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 2 \cdot 2 = 4$.

Let $\gamma = \sqrt{2}+\sqrt{3}$. Then $\gamma^{-1} = \frac{1}{\sqrt{3}+\sqrt{2}} = \sqrt{3}-\sqrt{2}$, so $\sqrt{3} = \frac{\gamma+\gamma^{-1}}{2}$ and $\sqrt{2} = \frac{\gamma-\gamma^{-1}}{2}$ lie in $\mathbb{Q}(\gamma)$. Hence $\mathbb{Q}(\gamma) = \mathbb{Q}(\sqrt{2},\sqrt{3})$ has degree 4. Now $\gamma^2 = 5+2\sqrt{6}$, so $(\gamma^2-5)^2 = 24$, i.e. $\gamma^4 - 10\gamma^2 + 1 = 0$. This monic polynomial has degree $4 = [\mathbb{Q}(\gamma) : \mathbb{Q}]$, so it is the minimal polynomial: $m_\gamma = x^4 - 10x^2 + 1$.

5.

$72 = 2^3 \cdot 3^2$. Partitions of 3: $3, 2+1, 1+1+1$; of 2: $2, 1+1$. So $3 \times 2 = 6$ groups:

Elementary divisors	Invariant factors
$8, 9$	$\mathbb{Z}/72$
$8, 3, 3$	$\mathbb{Z}/3 \oplus \mathbb{Z}/24$
$4, 2, 9$	$\mathbb{Z}/2 \oplus \mathbb{Z}/36$
$4, 2, 3, 3$	$\mathbb{Z}/6 \oplus \mathbb{Z}/12$
$2, 2, 2, 9$	$\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/18$
$2, 2, 2, 3, 3$	$\mathbb{Z}/2 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/6$

(To convert, multiply the largest power of each prime to get the largest invariant factor, then the next largest, and so on.)

6.

$x^2+x+1 \in \mathbb{F}_2[x]$ has no roots in $\mathbb{F}_2$, so (being of degree 2) it is irreducible, and $\mathbb{F}_4 := \mathbb{F}_2[x]/(x^2+x+1)$ is a field with basis $1, \alpha$ (α the class of x), so 4 elements $0, 1, \alpha, \alpha+1$, with $\alpha^2 = \alpha+1$.

$\times$	1	α	$\alpha+1$
1	1	α	$\alpha+1$
α	α	$\alpha+1$	1
$\alpha+1$	$\alpha+1$	1	α

(E.g. $\alpha(\alpha+1) = \alpha^2+\alpha = 2\alpha+1 = 1$, and $(\alpha+1)^2 = \alpha^2+1 = \alpha$.) So $\mathbb{F}_4^\times = \{1, \alpha, \alpha^2\}$ is cyclic of order 3, generated by α.

Core

7.

Let $P = (p) \neq 0$ be prime, and suppose $P \subseteq M = (m)$. Then $p = mr$ for some r. As p is prime, $p \mid m$ or $p \mid r$. If $p \mid m$, then $M \subseteq P$, so $M = P$. If $r = ps$, then $p = mps$, and cancelling $p \neq 0$ (domain) gives $ms = 1$, so $M = R$. Hence P is maximal.

In $F[x,y]$, (x) is a non-zero prime ideal ($F[x,y]/(x) \cong F[y]$ is a domain) which is not maximal: $(x) \subsetneq (x,y) \subsetneq F[x,y]$ (as $F[x,y]/(x,y) \cong F$). So $F[x,y]$ is not a PID.

8.

Use integer row and column operations (these replace A by PAQ with P, Q invertible over $\mathbb{Z}$, which does not change $\mathbb{Z}^3/A\mathbb{Z}^3$ up to isomorphism).

$C_2 \to C_2 - 2C_1$, $C_3 \to C_3 - 2C_1$:

$$\begin{pmatrix} 2 & 0 & 0 \\ -6 & 18 & 24 \\ 10 & -24 & -36 \end{pmatrix}$$

$R_2 \to R_2 + 3R_1$, $R_3 \to R_3 - 5R_1$:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 18 & 24 \\ 0 & -24 & -36 \end{pmatrix}$$

On the lower block $B = \begin{pmatrix} 18 & 24 \\ -24 & -36 \end{pmatrix}$: $R_3 \to R_3 + R_2$ gives $\begin{pmatrix} 18 & 24 \\ -6 & -12 \end{pmatrix}$; swap the rows and negate the new first row: $\begin{pmatrix} 6 & 12 \\ 18 & 24 \end{pmatrix}$; $C_3 \to C_3 - 2C_2$: $\begin{pmatrix} 6 & 0 \\ 18 & -12 \end{pmatrix}$; $R_3 \to R_3 - 3R_2$: $\begin{pmatrix} 6 & 0 \\ 0 & -12 \end{pmatrix}$; negate: $\operatorname{diag}(6, 12)$.

So the Smith normal form is $\operatorname{diag}(2, 6, 12)$ (and $2 \mid 6 \mid 12$). Check: $|\det A| = 144 = 2 \cdot 6 \cdot 12$. Therefore

$$G \cong \mathbb{Z}/2 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/12, \quad |G| = 144.$$

(Elementary divisors: $2, 2, 4, 3, 3$.)

9.

(a) For eigenvalue 2: block sizes sum to 4 (exponent in χ) with largest block 2 (exponent in m): partitions $2+2$ or $2+1+1$. For eigenvalue 3: sizes sum to 2, largest 1: $1+1$. So two Jordan forms:

- $J_2(2) \oplus J_2(2) \oplus J_1(3) \oplus J_1(3)$. Elementary divisors $(x-2)^2, (x-2)^2, (x-3), (x-3)$. Invariant factors: $(x-2)^2(x-3)$, $(x-2)^2(x-3)$.
- $J_2(2) \oplus J_1(2) \oplus J_1(2) \oplus J_1(3) \oplus J_1(3)$. Elementary divisors $(x-2)^2, (x-2), (x-2), (x-3), (x-3)$. Invariant factors: $(x-2)$, $(x-2)(x-3)$, $(x-2)^2(x-3)$.

(In each case the largest invariant factor is m and the product is χ, as it must be. The rational canonical form is the block diagonal matrix of the companion matrices of the invariant factors.)

(b) Matrices are similar iff they have the same invariant factors $a_1 \mid \cdots \mid a_k$ (monic, of degree ≥ 1), where $a_k = m$ and $a_1 \cdots a_k = \chi$. For $n = 3$:

- If $\deg m = 3$: $m = \chi$ and the list is just (m).
- If $\deg m = 2$: the remaining factors have product χ/m of degree 1, so the list is $(\chi/m, m)$.
- If $\deg m = 1$: every a_i divides m and has degree ≥ 1, so each $a_i = m$, and the list is (m, m, m).

In every case the invariant factors are determined by χ and m, so the matrices are similar.

For $n = 4$: $N_1 = J_2(0) \oplus J_2(0)$ and $N_2 = J_2(0) \oplus J_1(0) \oplus J_1(0)$ both have $\chi = x^4$ and $m = x^2$, but $\operatorname{rank} N_1 = 2 \neq 1 = \operatorname{rank} N_2$, so they are not similar.

10.

A satisfies $x^m - 1$, so its minimal polynomial divides $x^m - 1$. Over $\mathbb{C}$, $x^m - 1 = \prod_{k=0}^{m-1}(x - e^{2\pi i k/m})$ has m distinct roots. So m_A splits into distinct linear factors, hence A is diagonalisable (the elementary divisors are all linear, so all Jordan blocks have size 1).

Over $\mathbb{F}_p$: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has $A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$, so $A^p = I$. Its only eigenvalue is 1, so if it were diagonalisable it would equal I. It is not. (Here $x^p - 1 = (x-1)^p$ has a repeated root.)

11.

(a) f is irreducible (Problem 3(b)), so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ and $1, \alpha, \alpha^2$ is a basis. Using $\alpha^3 = 3\alpha + 1$: $\alpha^4 = 3\alpha^2 + \alpha$ and $\alpha^6 = (3\alpha+1)^2 = 9\alpha^2 + 6\alpha + 1$. Let $\beta = 2 - \alpha^2$. Then

$$\beta^3 = 8 - 12\alpha^2 + 6\alpha^4 - \alpha^6 = 8 - 12\alpha^2 + 18\alpha^2 + 6\alpha - 9\alpha^2 - 6\alpha - 1 = 7 - 3\alpha^2,$$

so $\beta^3 - 3\beta - 1 = 7 - 3\alpha^2 - 6 + 3\alpha^2 - 1 = 0$. Also $\beta \neq \alpha$: otherwise $\alpha^2 + \alpha - 2 = (\alpha+2)(\alpha-1) = 0$, making α rational. The three roots sum to 0 (no x^2 term), so the third root is $-\alpha - \beta = \alpha^2 - \alpha - 2 \in \mathbb{Q}(\alpha)$, and it differs from α and β since f is irreducible over $\mathbb{Q}$, hence separable, so has distinct roots. Thus $\mathbb{Q}(\alpha)$ contains all roots and is the splitting field.

(b) $\mathbb{Q}(\alpha)/\mathbb{Q}$ is a splitting field of a separable polynomial, hence Galois, of degree 3. So the Galois group has order 3: it is $\mathbb{Z}/3 \cong A_3$, generated by $\alpha \mapsto 2 - \alpha^2$.

Check: for $x^3 + px + q$, $\Delta = -4p^3 - 27q^2$. Here $p = -3$, $q = -1$: $\Delta = 108 - 27 = 81 = 9^2$, a square in $\mathbb{Q}$, so $\mathrm{Gal}(f) \leq A_3$; since f is irreducible, the group is transitive on 3 roots, so it is A_3.

12.

G is a finite abelian group, so by the structure theorem $G \cong \mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$ with $1 < d_1 \mid \cdots \mid d_k$. Then every $g \in G$ satisfies $g^{d_k} = 1$, i.e. all $|G| = d_1 \cdots d_k$ elements are roots of $x^{d_k} - 1$. A polynomial of degree d_k over a field has at most d_k roots, so $d_1 \cdots d_k \leq d_k$, forcing $k = 1$ (or G trivial). So $G \cong \mathbb{Z}/d_k$ is cyclic.

13.

If $E \subseteq \mathbb{F}_{p^n}$ is a subfield with p^m elements, then $[E : \mathbb{F}_p] = m$ and by the tower law $n = m\,[\mathbb{F}_{p^n} : E]$, so $m \mid n$.

Conversely, let $m \mid n$, say $n = mk$. Then $p^m - 1$ divides $p^n - 1 = (p^m)^k - 1$, and hence $x^{p^m-1} - 1$ divides $x^{p^n-1} - 1$ (if $a \mid b$ then $y - 1 \mid y^{b/a} - 1$ with $y = x^a$). Multiplying by x: $x^{p^m} - x$ divides $x^{p^n} - x$. All p^n elements of $\mathbb{F}_{p^n}$ are roots of $x^{p^n} - x$, which therefore splits into distinct linear factors over $\mathbb{F}_{p^n}$; so does its factor $x^{p^m} - x$. Let E be its set of roots in $\mathbb{F}_{p^n}$; $|E| = p^m$ (distinct roots since the derivative is -1). E is a subfield: it contains $0, 1$, and if $a^{p^m} = a$, $b^{p^m} = b$ then $(a \pm b)^{p^m} = a \pm b$ (Frobenius is additive), $(ab)^{p^m} = ab$, and $(a^{-1})^{p^m} = a^{-1}$.

Uniqueness: every element of a subfield E' with p^m elements satisfies $a^{p^m} = a$ (Lagrange in $E'^{\times}$, plus $a = 0$), so $E' \subseteq E$, and equality holds by counting.

14.

(a) Let $P = x^{p^n} - x$. Since $P' = -1$, $\gcd(P, P') = 1$, so P is squarefree: it is the product of the distinct monic irreducibles dividing it. It remains to show that a monic irreducible g of degree d divides P iff $d \mid n$.

If $d \mid n$: $L = \mathbb{F}_p[x]/(g)$ is a field with p^d elements, and the class α of x satisfies $\alpha^{p^d} = \alpha$. So $g = m_\alpha$ divides $x^{p^d} - x$, which divides P (Problem 13's argument, since $d \mid n$).

If $g \mid P$: P splits in $\mathbb{F}_{p^n}$ (its roots are all the elements), so g has a root $\alpha \in \mathbb{F}_{p^n}$, and $\mathbb{F}_p(\alpha)$ is a subfield of degree d. By Problem 13 (or the tower law), $d \mid n$.

(b) Comparing degrees in (a): $p^n = \sum_{d|n} dN_d$. For $p = 2$: $n = 1$: $2 = N_1$. $n = 2$: $4 = N_1 + 2N_2$, so $N_2 = 1$. $n = 4$: $16 = N_1 + 2N_2 + 4N_4 = 4 + 4N_4$, so $N_4 = 3$.

To list them: a quartic over $\mathbb{F}_2$ with no root has constant term 1 and an odd number of non-zero terms. The candidates are x^4+x+1, x^4+x^3+1, x^4+x^2+1, $x^4+x^3+x^2+x+1$. A rootless reducible quartic is a product of two irreducible quadratics, which must be $(x^2+x+1)^2 = x^4+x^2+1$. So the irreducible quartics are

$$x^4 + x + 1, \quad x^4 + x^3 + 1, \quad x^4 + x^3 + x^2 + x + 1.$$

15.

(a) Let f be irreducible of degree $n \geq 1$. In characteristic 0, f' has degree $n-1$ and leading coefficient $n \cdot (\text{lead} f) \neq 0$, so $f' \neq 0$. $\gcd(f, f')$ divides the irreducible f, so it is 1 or (an associate of) f; it cannot be f since $f \nmid f'$ for degree reasons. So $\gcd(f, f') = 1$ and f is separable.

(b) t is a prime element of the UFD $\mathbb{F}_p[t]$, and $x^p - t$ is Eisenstein at t (all non-leading coefficients, namely $-t$, are divisible by t, the constant is not divisible by t^2, and the leading coefficient 1 is not divisible by t). By Eisenstein and Gauss's lemma it is irreducible in $\mathbb{F}_p(t)[x]$. In a splitting field let s be a root, so $s^p = t$; then $x^p - t = x^p - s^p = (x-s)^p$ (characteristic p). So it has only one distinct root: not separable. (Equivalently, its derivative px^{p-1} is 0.)

16.

(a) ζ is a root of $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$, and

$$\Phi_5(x+1) = \frac{(x+1)^5 - 1}{x} = x^4 + 5x^3 + 10x^2 + 10x + 5$$

is Eisenstein at 5. So Φ_5 is irreducible and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$. The roots $\zeta, \zeta^2, \zeta^3, \zeta^4$ all lie in $\mathbb{Q}(\zeta)$, so it is a splitting field (of a separable polynomial), hence Galois. Each σ in the Galois group sends ζ to some ζ^a, $a \in (\mathbb{Z}/5)^\times$, and $\sigma \mapsto a$ is an injective homomorphism (as $\sigma_a \sigma_b(\zeta) = \zeta^{ab}$). Both groups have order 4, so Gal $\cong (\mathbb{Z}/5)^\times$, which is cyclic, generated by 2 $(2, 4, 3, 1)$.

(b) A cyclic group of order 4 has exactly one subgroup of order 2: $H = \{\sigma_1, \sigma_4\}$ (σ_4 is complex conjugation). By the fundamental theorem, $E = \mathbb{Q}(\zeta)^H$ is the unique quadratic subfield. $\eta = \zeta + \zeta^4$ is fixed by σ_4, so $\eta \in E$. Let $\eta' = \zeta^2 + \zeta^3$. Then $\eta + \eta' = -1$ and

$$\eta\eta' = \zeta^3 + \zeta^4 + \zeta^6 + \zeta^7 = \zeta^3 + \zeta^4 + \zeta + \zeta^2 = -1.$$

So η is a root of $x^2 + x - 1$, i.e. $\eta = \frac{-1 \pm \sqrt{5}}{2}$ (in fact $\eta = 2\cos 72^\circ = \frac{-1+\sqrt{5}}{2}$). Since $\eta \notin \mathbb{Q}$, $\mathbb{Q}(\eta) = \mathbb{Q}(\sqrt{5})$ is a quadratic subfield of E, hence $E = \mathbb{Q}(\sqrt{5})$.

17.

Let $\beta = \sqrt[4]{2} > 0$. The roots of $x^4 - 2$ are $\pm\beta, \pm i\beta$, so the splitting field is $K = \mathbb{Q}(\beta, i\beta) = \mathbb{Q}(\beta, i)$. $x^4 - 2$ is Eisenstein at 2, so $[\mathbb{Q}(\beta) : \mathbb{Q}] = 4$. Since $\mathbb{Q}(\beta) \subseteq \mathbb{R}$, $i \notin \mathbb{Q}(\beta)$, so $[K : \mathbb{Q}(\beta)] = 2$ and $[K : \mathbb{Q}] = 8$. Hence also $[K : \mathbb{Q}(i)] = 4$, so $x^4 - 2$ is irreducible over $\mathbb{Q}(i)$.

$K/\mathbb{Q}$ is Galois (splitting field, characteristic 0), so $|G| = 8$. Any $\sigma \in G$ is determined by $\sigma(\beta) \in \{\pm\beta, \pm i\beta\}$ and $\sigma(i) \in \{\pm i\}$.

- Since $x^4 - 2$ is irreducible over $\mathbb{Q}(i)$ and K is its splitting field over $\mathbb{Q}(i)$, there is $r \in \mathrm{Gal}(K/\mathbb{Q}(i))$ with $r(\beta) = i\beta$, $r(i) = i$.
- Complex conjugation restricts to $s \in G$ with $s(\beta) = \beta$, $s(i) = -i$.

Then $r^2(\beta) = r(i\beta) = i \cdot i\beta = -\beta$, $r^3(\beta) = -i\beta$, $r^4 = 1$: r has order 4. s has order 2, $s \notin \langle r \rangle$ (elements of $\langle r \rangle$ fix i). And

$$srs(\beta) = sr(\beta) = s(i\beta) = -i\beta = r^3(\beta), \qquad srs(i) = sr(-i) = s(-i) = i = r^3(i),$$

so $srs = r^{-1}$. Thus $G = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$, the dihedral group of order 8.

18.

φ is a ring homomorphism ($(ab)^p = a^p b^p$; $(a+b)^p = a^p + b^p$ since $p \mid \binom{p}{k}$ for $0 < k < p$). It is injective (field homomorphism), hence bijective on the finite set $\mathbb{F}_{p^n}$. It fixes $\mathbb{F}_p$ by Fermat's little theorem. So $\varphi \in \mathrm{Aut}(\mathbb{F}_{p^n}/\mathbb{F}_p)$.

If $\varphi^k = \mathrm{id}$ with $1 \leq k$, then every element is a root of $x^{p^k} - x$, so $p^n \leq p^k$, i.e. $k \geq n$. And $\varphi^n = \mathrm{id}$ since $a^{p^n} = a$ for all a. So φ has order exactly n. Since $|\mathrm{Aut}(\mathbb{F}_{p^n}/\mathbb{F}_p)| \leq [\mathbb{F}_{p^n} : \mathbb{F}_p] = n$, the automorphism group is $\langle \varphi \rangle$, cyclic of order $n = [\mathbb{F}_{p^n} : \mathbb{F}_p]$. So the extension is Galois.

19.

Suppose $\theta : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(\sqrt{3})$ is a field isomorphism. It fixes 1, hence $\mathbb{Z}$, hence $\mathbb{Q}$. Then $\theta(\sqrt{2})^2 = \theta(2) = 2$, so $\mathbb{Q}(\sqrt{3})$ contains a square root of 2, say $a + b\sqrt{3}$ with $a, b \in \mathbb{Q}$. Squaring: $a^2 + 3b^2 + 2ab\sqrt{3} = 2$, so $ab = 0$. If $b = 0$, $a^2 = 2$; if $a = 0$, $3b^2 = 2$, i.e. $(3b)^2 = 6$. Neither has a rational solution. Contradiction.

Challenge

20.

Irreducible: Eisenstein at $p = 3$.

Exactly three real roots: $f(-2) = -17$, $f(-1) = 8$, $f(1) = -2$, $f(2) = 23$. By the intermediate value theorem there are roots in $(-2, -1)$, $(-1, 1)$ and $(1, 2)$. On the other hand $f'(x) = 5x^4 - 6$ has exactly two real zeros, so by Rolle's theorem f has at most three real roots. So f has exactly three real roots and two non-real roots, which are complex conjugates.

The Galois group. Let $K \subseteq \mathbb{C}$ be the splitting field and $G = \mathrm{Gal}(K/\mathbb{Q})$, acting faithfully on the 5 roots, so $G \leq S_5$.

- If α is a root, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 5$ divides $[K : \mathbb{Q}] = |G|$. By Cauchy's theorem G has an element of order 5, which in S_5 must be a 5-cycle.
- Complex conjugation maps K to itself (it permutes the roots of f), so restricts to an element of G. It fixes the three real roots and swaps the two non-real ones: a transposition.

Lemma: a p-cycle σ and a transposition τ generate S_p (p prime). Write $\tau = (a\ b)$. Some power σ^k sends a to b, and since p is prime, σ^k is again a p-cycle. Relabel the points so that $\sigma^k = (1\ 2\ \cdots\ p)$ and $\tau = (1\ 2)$. Then $\sigma^{kj}\tau\sigma^{-kj} = (1+j\ \ 2+j)$ for $j = 0, \ldots, p-2$. These adjacent transpositions generate S_p (every permutation is a product of adjacent transpositions, as in bubble sort).

So $G = S_5$. S_5 is not solvable: its subgroup A_5 is simple and non-abelian, and subgroups of solvable groups are solvable. By Galois's theorem (characteristic 0: solvable by radicals $\Leftrightarrow$ solvable Galois group), the roots of f cannot be expressed by radicals.

21.

Use $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$; with $u = 2\cos\theta$ this reads $u^3 - 3u = 2\cos 3\theta$.

(a) The angle 60° is constructible (equilateral triangle). If it could be trisected, 20° and hence $\cos 20^\circ$ and $u = 2\cos 20^\circ$ would be constructible. But $u^3 - 3u = 2\cos 60^\circ = 1$, so u is a root of $x^3 - 3x - 1$, which is irreducible (Problem 3(b)). So $[\mathbb{Q}(u) : \mathbb{Q}] = 3$, not a power of 2. Contradiction.

(b) Constructing a regular 9-gon would construct the central angle 40°, hence $u = 2\cos 40^\circ$. Now $u^3 - 3u = 2\cos 120^\circ = -1$, so u is a root of $x^3 - 3x + 1$. Its only possible rational roots are ± 1, and $1 - 3 + 1 = -1$, $-1 + 3 + 1 = 3$, so it is irreducible, and $[\mathbb{Q}(u) : \mathbb{Q}] = 3$. Contradiction.

22.

If $a = 0$, $K = F$ and there is nothing to prove; assume $a \neq 0$. The roots of $x^n - a$ are $\zeta^k\beta$, $k = 0, \dots, n-1$; they are distinct (as $\beta \neq 0$ and ζ has order n) and all lie in K since $\zeta \in F$. So K is the splitting field over F of the separable polynomial $x^n - a$, and K/F is Galois.

For $\sigma \in G = \mathrm{Gal}(K/F)$, $\sigma(\beta)$ is a root of $x^n - a$, so $\sigma(\beta) = \omega_\sigma\beta$ for a unique $\omega_\sigma \in \mu_n = \langle\zeta\rangle$. Since σ fixes $F \ni \zeta$,

$$\sigma\tau(\beta) = \sigma(\omega_\tau\beta) = \omega_\tau\,\omega_\sigma\beta,$$

so $\sigma \mapsto \omega_\sigma$ is a homomorphism $G \to \mu_n$. It is injective because σ is determined by $\sigma(\beta)$ ($K = F(\beta)$). Hence G is isomorphic to a subgroup of the cyclic group $\mu_n \cong \mathbb{Z}/n$, so G is cyclic of order dividing n.

23.

Let $f = m_\alpha$, $g = m_\beta$ over F, and let L be a splitting field of fg over K. Let $\alpha = \alpha_1, \dots, \alpha_r$ be the roots of f and $\beta = \beta_1, \dots, \beta_s$ the roots of g in L; by separability the β_j are distinct. For $j \neq 1$, $\beta_j \neq \beta$, so the equation $\alpha_i + c\beta_j = \alpha + c\beta$ has at most one solution $c = (\alpha_i - \alpha)/(\beta - \beta_j)$. Exclude these finitely many values; since F is infinite, all other $c \in F$ remain. Fix such a c and put $\gamma = \alpha + c\beta \in K$.

Let $h(x) = f(\gamma - cx) \in F(\gamma)[x]$. Then $h(\beta) = f(\alpha) = 0$, and for $j \neq 1$, $h(\beta_j) = f(\gamma - c\beta_j) \neq 0$ because $\gamma - c\beta_j \neq \alpha_i$ for all i by the choice of c. So in $L[x]$, the only common root of g and h is β, and it is a simple root of g. Therefore $\gcd(g, h) = x - \beta$ in $L[x]$. The gcd of two polynomials in $F(\gamma)[x]$ can be computed by the Euclidean algorithm inside $F(\gamma)[x]$, so it is the same whether computed in $F(\gamma)[x]$ or $L[x]$. Hence $x - \beta \in F(\gamma)[x]$, i.e. $\beta \in F(\gamma)$, and then $\alpha = \gamma - c\beta \in F(\gamma)$. So $K = F(\alpha, \beta) = F(\gamma)$.

For a general finite separable extension $K = F(\alpha_1, \dots, \alpha_n)$ (finitely many generators suffice, e.g. a basis), induct: $F(\alpha_1, \alpha_2) = F(\gamma_2)$, then $F(\gamma_2, \alpha_3) = F(\gamma_3)$, etc. (Each intermediate field is separable over F, as a subfield of K.)

24.

(a) By the structure theorem, a finitely generated R-module is $M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k)$ with each d_i a non-zero non-unit. The element $(0, \dots, 0, 1, 0, \dots)$ with 1 in the $R/(d_i)$ slot is non-zero and killed by $d_i \neq 0$, i.e. it is torsion. So if M is torsion-free, $k = 0$ and $M \cong R^r$ is free.

(b) $\mathbb{Q}$ is torsion-free: $nq = 0$ with $n \neq 0$ forces $q = 0$. Suppose $\mathbb{Q}$ were free with basis B. B has at most one element: for any $p/q, r/s \in \mathbb{Q}$ (non-zero),

$$(qr) \cdot \frac{p}{q} - (ps) \cdot \frac{r}{s} = pr - pr = 0$$

is a non-trivial $\mathbb{Z}$-linear relation. And $B \neq \emptyset$ since $\mathbb{Q} \neq 0$. So $\mathbb{Q} = \mathbb{Z}b$ for some b. But then $b/2 \in \mathbb{Q}$ equals nb for some $n \in \mathbb{Z}$, giving $(2n-1)b = 0$, so $b = 0$: contradiction. Hence $\mathbb{Q}$ is not free.

By (a), $\mathbb{Q}$ is therefore **not finitely generated** as a $\mathbb{Z}$-module. (Direct check: finitely many fractions have a common denominator D, and $1/(2D)$ is not in the subgroup they generate.)